



COEP Technological University

(COEP Tech)

A Unitary Public University of Government of Maharashtra

Mid Semester Examination

Program : S.Y. B.Tech.

Semester: III

Course Code: OEC-24016

Course Name: Complex Analysis

Branch: AL

Academic Year: 2025-26

Duration: 90 mins

Maximum Marks: 30

Date : / /2026

Student MIS Number :

--	--	--	--	--	--	--	--	--	--

Instructions :

1. Write your MIS Number and Branch on the Question Paper.
2. Writing anything else on the question paper is not allowed.
3. Read a given question very carefully and then answer.
4. Mobile phones and calculators are strictly prohibited.
5. Exchange/Sharing of stationery, etc. is not allowed.
6. Figures to the right indicate the course outcomes, full marks and Learning level.
7. Any essential result, formula or theorem assumed for answering questions must be clearly stated.
8. Let $\mathbb{C}$ denotes the set of complex numbers.
9. For $z \in \mathbb{C} \setminus \{0\}$, let $\arg(z)$ be chosen such that $\arg(z) \in (-\pi, \pi]$.

1. State True or False **with justification**.

- (a) There exists a set $S \subset \mathbb{C}$ which is neither open but not closed. [CO2, 1 Mark, L2]
- (b) Let A and B be two compact sets in $\mathbb{C}$. Then $A \cup B$ is a compact set. [CO2, 1 Mark, L2]
- (c) We have $(1 + i)^4 = 2$. [CO1, 1 Mark, L1]
- (d) The set $\{z \in \mathbb{C} : z = |\bar{z}|^2\}$ is not bounded. [CO3, 1 Mark, L3]
- (e) Let A be an open set in $\mathbb{C}$. Let $B \subset A$, then A is an open set in $\mathbb{C}$. [CO1, 1 Mark, L1]
- (f) Let $z_1, z_2 \in \mathbb{C}$, both non zero, then $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$. [CO3, 1 Mark, L3]

2. Let $g : \mathbb{C} \rightarrow \mathbb{C}$ be given by $g(z) = |z - 1| - \frac{z+\bar{z}}{2}$.

- (a) Tell if g is continuous on $\mathbb{C}$ with justification. [CO3, 2 Marks, L3]
- (b) Let $E = \{z : |z - 1| = \frac{z+\bar{z}}{2}\}$. Show that E is a closed set. [CO3, 1 Marks, L3]
- (c) Check if g is complex differentiable at 0
without using Cauchy Riemann equation. [CO3, 1 Marks, L3]

3. Find the following limits whenever the limit exists. If the limit does not exist, then justify its nonexistence. **(SOLVE ANY TWO)** [CO3/4, (2+2) Marks, L3/4]

(a)

$$\lim_{n \rightarrow \infty} \frac{n^2 + 3in + \sin n + e^{-n}}{2in^2 - 5n + \cos n}.$$

(b)

$$\lim_{z \rightarrow 3i} \frac{(\cos z)(z^3 + 27i)}{z^2 + 9}$$

(c)

$$\lim_{z \rightarrow 0} \frac{xy^2}{x^3 + xy^2 + y^3} + i \frac{x^2}{x + y},$$

where $z = x + iy$.

4. Let z_n be a sequence of complex numbers converging to z_0 , that is,

$$\lim_{n \rightarrow \infty} z_n = z_0.$$

(a) Prove that $|z_n|$ converges to $|z_0|$. [CO3, 1 Marks, L1]

(b) Give an example of a sequence z_n such that $\lim_{n \rightarrow \infty} |z_n| = |z_0|$ but $\lim_{n \rightarrow \infty} z_n \neq z_0$. [CO3, 1 Marks, L3]

(c) Is it possible to give a sequence z_n such that $\lim_{n \rightarrow \infty} |z_n| = 0$ but $\lim_{n \rightarrow \infty} z_n \neq 0$? Justify. [CO2, 1 Marks, L2]

(d) If the sequence z_n converges to z_0 , then prove that the sequence

$$s_n = \frac{1}{n} \sum_{k=1}^n z_k$$

also converges to z_0 . [CO4, 3 Marks, L4]

5. Let $f : D \rightarrow \mathbb{C}$, where D is a open set in $\mathbb{C}$. For $z \in D$, let $f(z) = u(z) + iv(z)$, where u and v are real valued functions on D . Let $\text{Re}(f) : D \rightarrow \mathbb{R}$ and $\text{Im}(f) : D \rightarrow \mathbb{R}$ be given by $\text{Re}(f)(z) = u(z)$, $\text{Im}(f)(z) = v(z)$.

Show that f is continuous in D

if and only if both $\text{Re}(f)$ and $\text{Im}(f)$ are continuous in D . [CO4, 3 Marks, L3]

6. (a) Find two complex numbers z such that $\cos z = 4$. [CO3, 2 Marks, L3]

(b) Prove that $\sin^2 z + \cos^2 z = 1$, where $z \in \mathbb{C}$. [CO3, 2 Marks, L3]

7. Let $f(z) = \bar{z}^2$, where $z \in \mathbb{C}$. Find all $z \in \mathbb{C}$ where f does not satisfy the Cauchy Riemann equation. Find all points where f is complex differentiable. [CO3, 3 Marks, L3]

OR

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be given by $f(z) = e^x \cos(ay) + ie^x \sin(y + b) + c$. Find restrictions on the constants a, b, c so that f is complex differentiable on $\mathbb{C}$.

Mid Semester Examination – Complex Analysis

1. True or False with justification

- (a) **True.** Take, for example, $S = [0, 1) \subset \mathbb{C}$ (viewing $\mathbb{R} \subset \mathbb{C}$). Then S is not open in $\mathbb{C}$, and it is not closed since 1 is a limit point of S but $1 \notin S$.
- (b) **True.** A finite union of compact sets is compact. Since A and B are compact in $\mathbb{C}$, their union $A \cup B$ is compact.

- (c) **False.**

$$(1+i)^2 = 1 + 2i + i^2 = 2i, \quad (1+i)^4 = (2i)^2 = -4.$$

Hence $(1+i)^4 \neq 2$.

- (d) **False.** Since $|\bar{z}| = |z|$, the set becomes

$$\{z \in \mathbb{C} : z = |z|^2\}.$$

If $z = x + iy$, then the right-hand side is real, so $y = 0$. Thus $z = x \in \mathbb{R}$ and

$$x = x^2 \implies x = 0 \text{ or } 1.$$

So the set is $\{0, 1\}$, which is bounded.

- (e) **False.** If A is open in $\mathbb{C}$, then A is open regardless of $B \subset A$. The statement as written is logically incorrect because $B \subset A$ does not imply anything new about openness of A . If the intended statement was “ B is open,” that would be false in general.
- (f) **False.** In general,

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

need not hold exactly because the principal argument must be taken modulo 2π . The correct relation is

$$\arg(z_1 z_2) \equiv \arg(z_1) + \arg(z_2) \pmod{2\pi}.$$

2. Let $g(z) = |z - 1| - \frac{z + \bar{z}}{2}$

- (a) g is continuous on $\mathbb{C}$. The function $z \mapsto |z - 1|$ is continuous on $\mathbb{C}$, and

$$\frac{z + \bar{z}}{2} = \operatorname{Re}(z)$$

is also continuous on $\mathbb{C}$. Therefore their difference $g(z)$ is continuous on $\mathbb{C}$.

- (b) Let

$$E = \left\{ z \in \mathbb{C} : |z - 1| = \frac{z + \bar{z}}{2} \right\}.$$

Define

$$h(z) = |z - 1| - \frac{z + \bar{z}}{2}.$$

Then h is continuous on $\mathbb{C}$, and

$$E = h^{-1}(\{0\}).$$

Since $\{0\}$ is closed in $\mathbb{C}$, the inverse image of a closed set under a continuous function is closed. Hence E is closed.

(c) We check complex differentiability at 0 directly from the definition:

$$g'(0) = \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h},$$

if this limit exists.

Now

$$g(0) = |-1| - \frac{0 + \bar{0}}{2} = 1.$$

Let $h \in \mathbb{C}$. Then

$$g(h) - g(0) = |h - 1| - \frac{h + \bar{h}}{2} - 1.$$

Take $h = t \in \mathbb{R}$, $t \rightarrow 0$. Then

$$\frac{g(t) - g(0)}{t} = \frac{|t - 1| - t - 1}{t}.$$

For small real t , $|t - 1| = 1 - t$, so

$$\frac{g(t) - g(0)}{t} = \frac{(1 - t) - t - 1}{t} = \frac{-2t}{t} = -2.$$

Now take $h = it$, $t \rightarrow 0$. Then

$$\frac{g(it) - g(0)}{it} = \frac{|it - 1| - \frac{it - it}{2} - 1}{it} = \frac{\sqrt{1 + t^2} - 1}{it}.$$

Since $\sqrt{1 + t^2} - 1 \sim \frac{t^2}{2}$, this tends to 0. Since the limits along two paths are different, $g'(0)$ does not exist. Thus g is not complex differentiable at 0.

3. Limits

(a)

$$\lim_{n \rightarrow \infty} \frac{n^2 + 3in + \sin n + e^{-n}}{2in^2 - 5n + \cos n}.$$

Divide numerator and denominator by n^2 :

$$\frac{1 + \frac{3i}{n} + \frac{\sin n}{n^2} + \frac{e^{-n}}{n^2}}{2i - \frac{5}{n} + \frac{\cos n}{n^2}}.$$

As $n \rightarrow \infty$, this tends to

$$\frac{1}{2i} = -\frac{i}{2}.$$

Hence the limit is

$$\boxed{-\frac{i}{2}}.$$

(b)

$$\lim_{z \rightarrow 3i} \frac{(\cos z)(z^3 + 27i)}{z^2 + 9}.$$

Factor:

$$z^3 + 27i = (z - 3i)(z^2 + 3iz - 9), \quad z^2 + 9 = (z - 3i)(z + 3i).$$

So for $z \neq 3i$,

$$\frac{(\cos z)(z^3 + 27i)}{z^2 + 9} = (\cos z) \frac{z^2 + 3iz - 9}{z + 3i}.$$

Now substitute $z = 3i$:

$$\cos(3i) \cdot \frac{(3i)^2 + 3i(3i) - 9}{3i + 3i} = \cos(3i) \cdot \frac{-9 - 9 - 9}{6i} = \cos(3i) \cdot \frac{-27}{6i}.$$

Since $\cos(3i) = \cosh 3$, the limit is

$$\boxed{\frac{9i}{2} \cosh 3}.$$

(c)

$$\lim_{z \rightarrow 0} \left(\frac{xy^2}{x^3 + xy^2 + y^3} + i \frac{x^2}{x + y} \right), \quad z = x + iy.$$

The second term is not defined along $x + y = 0$, and the first term depends on the path.

Take $y = 0$. Then

$$\frac{xy^2}{x^3 + xy^2 + y^3} = 0, \quad i \frac{x^2}{x + y} = ix,$$

so the expression tends to 0.

Take $x = 0$. Then the first term is 0, and

$$i \frac{x^2}{x + y} = 0.$$

So again the limit is 0.

But if we take $y = x$, then

$$\frac{xy^2}{x^3 + xy^2 + y^3} = \frac{x \cdot x^2}{x^3 + x^3 + x^3} = \frac{1}{3},$$

while

$$i \frac{x^2}{x + y} = i \frac{x^2}{2x} = \frac{ix}{2} \rightarrow 0.$$

Hence the expression tends to $\frac{1}{3}$ along this path.

Since different paths give different values, the limit does not exist.

4. Sequence of complex numbers

Let $z_n \rightarrow z_0$.

(a) Since the modulus is continuous,

$$|z_n| \rightarrow |z_0|.$$

More explicitly, by the reverse triangle inequality,

$$||z_n| - |z_0|| \leq |z_n - z_0| \rightarrow 0.$$

(b) Example:

$$z_n = -1 \quad \text{for all } n.$$

Then $|z_n| = 1 \rightarrow 1$. If $z_0 = 1$, then $|z_0| = 1$, but $z_n \not\rightarrow z_0$ since $z_n = -1$ for all n .

(c) Yes, it is possible. For example,

$$z_n = e^{in}.$$

Then $|z_n| = 1$ for all n , so $|z_n| \not\rightarrow 0$. But if $|z_n| \rightarrow 0$, then by $|z_n| \rightarrow 0$ and

$$|z_n - 0| = |z_n|,$$

we must have $z_n \rightarrow 0$. So the answer is **No**, it is not possible.

(d) We prove that

$$s_n = \frac{1}{n} \sum_{k=1}^n z_k \rightarrow z_0.$$

Write

$$s_n - z_0 = \frac{1}{n} \sum_{k=1}^n (z_k - z_0).$$

Let $\varepsilon > 0$. Since $z_n \rightarrow z_0$, there exists N such that

$$|z_k - z_0| < \varepsilon \quad \text{for all } k \geq N.$$

Then for $n \geq N$,

$$|s_n - z_0| \leq \frac{1}{n} \sum_{k=1}^{N-1} |z_k - z_0| + \frac{1}{n} \sum_{k=N}^n |z_k - z_0|.$$

Let

$$M = \sum_{k=1}^{N-1} |z_k - z_0|.$$

Then

$$|s_n - z_0| \leq \frac{M}{n} + \varepsilon.$$

As $n \rightarrow \infty$, $\frac{M}{n} \rightarrow 0$, so

$$\limsup_{n \rightarrow \infty} |s_n - z_0| \leq \varepsilon.$$

Since ε is arbitrary, $s_n \rightarrow z_0$.

5. Continuity of f

Let

$$f(z) = u(z) + iv(z).$$

Then f is continuous at z_0 if and only if both u and v are continuous at z_0 .

Indeed, if f is continuous at z_0 , then

$$u(z) = \operatorname{Re}(f(z)), \quad v(z) = \operatorname{Im}(f(z)).$$

Since the real and imaginary part maps are continuous, u and v are continuous.

Conversely, if u and v are continuous at z_0 , then

$$f(z) - f(z_0) = (u(z) - u(z_0)) + i(v(z) - v(z_0)).$$

Hence

$$|f(z) - f(z_0)| \leq |u(z) - u(z_0)| + |v(z) - v(z_0)|.$$

Both terms on the right tend to 0, so f is continuous at z_0 . Therefore f is continuous in D if and only if $\operatorname{Re}(f)$ and $\operatorname{Im}(f)$ are continuous in D .

6. Trigonometric identities

(a) Find z such that $\cos z = 4$.

Let $z = x + iy$. Then

$$\cos z = \cos x \cosh y - i \sin x \sinh y.$$

For $\cos z$ to be real, we need

$$\sin x \sinh y = 0.$$

Take $x = 2k\pi$, so $\sin x = 0$. Then

$$\cos z = \cosh y.$$

We need $\cosh y = 4$, so

$$y = \pm \cosh^{-1}(4) = \pm \log(4 + \sqrt{15}).$$

Thus two complex numbers satisfying $\cos z = 4$ are

$$\boxed{z = i \log(4 + \sqrt{15}), \quad z = -i \log(4 + \sqrt{15}).}$$

(b) Prove that

$$\sin^2 z + \cos^2 z = 1.$$

Using the exponential forms,

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}.$$

Then

$$\sin^2 z + \cos^2 z = \frac{(e^{iz} - e^{-iz})^2}{-4} + \frac{(e^{iz} + e^{-iz})^2}{4}.$$

Simplifying,

$$\sin^2 z + \cos^2 z = \frac{e^{2iz} - 2 + e^{-2iz}}{4} + \frac{e^{2iz} + 2 + e^{-2iz}}{4} = 1.$$

Hence

$$\boxed{\sin^2 z + \cos^2 z = 1}.$$

7. Cauchy–Riemann equations

Let

$$f(z) = \bar{z}^2.$$

Write $z = x + iy$. Then

$$\bar{z} = x - iy, \quad f(z) = (x - iy)^2 = x^2 - y^2 - 2ixy.$$

So

$$u(x, y) = x^2 - y^2, \quad v(x, y) = -2xy.$$

Compute partial derivatives:

$$u_x = 2x, \quad u_y = -2y, \quad v_x = -2y, \quad v_y = -2x.$$

The Cauchy–Riemann equations require

$$u_x = v_y \quad \text{and} \quad u_y = -v_x.$$

Thus

$$2x = -2x \implies x = 0, \quad -2y = 2y \implies y = 0.$$

Therefore the Cauchy–Riemann equations hold only at $(0, 0)$, i.e. at $z = 0$.

However, complex differentiability at 0 must also be checked. We have

$$f'(0) = \lim_{h \rightarrow 0} \frac{\bar{h}^2 - 0}{h} = \lim_{h \rightarrow 0} \frac{\bar{h}^2}{h}.$$

Let $h = re^{i\theta}$. Then

$$\frac{\bar{h}^2}{h} = \frac{r^2 e^{-2i\theta}}{re^{i\theta}} = re^{-3i\theta} \rightarrow 0.$$

Hence f is complex differentiable at $z = 0$, and not differentiable anywhere else.

So:

f satisfies CR only at $z = 0$, and is complex differentiable only at $z = 0$.